Week 5

1) The primary goal of a consensus algorithm in distributed systems is to:
A. Maximize throughput
B. Ensure all non-faulty processes agree on the same value
C. Detect failures quickly
D. Synchronize physical clocks

2) Consensus is provably solvable in which system model?
A. Fully asynchronous with unbounded delays
B. Synchronous with bounded message and processing delays
C. Anonymous message-passing systems
D. Systems with Byzantine failures and no bounds

3) Paxos is best described as a protocol that is:
A. Unsafe but fast
B. Safe and eventually live
C. Fully synchronous
D. Byzantine-fault tolerant

4) In Paxos, who can initiate a new round of consensus?
A. Only the leader
B. Only the process with highest ID
C. Any process at any time
D. Only a majority quorum

5) Paxos may restart rounds repeatedly because:
A. Messages are always lost
B. The protocol violates safety
C. Leaders may fail or messages may be delayed
D. Acceptors refuse proposals

6) A Byzantine fault is characterized by a component that:
A. Stops execution permanently
B. Fails only by crashing
C. Behaves arbitrarily, possibly maliciously
D. Misses deadlines only

7) Which fault model is the most difficult to tolerate?
A. Crash fault
B. Omission fault
C. Fail-stop fault

D. Byzantine fault

8) An SLA (Service Level Agreement) consists of:
A. SLIs only
B. SLOs only
C. SLOs plus consequences for violation
D. Hardware guarantees only

9) The “domino effect” occurs when:
A. One rollback forces many other processes to roll back
B. Failures propagate through hardware
C. Messages are lost permanently
D. All checkpoints are coordinated

10) Agreement algorithms for Byzantine processes depend heavily on:
A. Faulty link detection
B. Message encryption only
C. Physical clock accuracy
D. Synchrony assumptions and failure models